# Statistical Methods 2: Discrete Random Variables

### Independence in Random Variables

Previously, we characterized disjoint events (where $P(A \cup B) = P(A) + P(B)$).

Disjoint events have mutually exclusive outcomes (you cannot roll a 4 and a 6 on the same die).

Alternatively, you can have independent events, where the probabilty of $A$ is unchanged given $B$:

Then, $P(A \mid B) = P(A)$, or equivalently $P(A \cap B) = P(A)P(B)$ (still defined when $P(B) = 0$).

**Recursive independence:** In general, a set of $n$ events is mutually independent if the probability of its intersection equals the product of its probabilities **AND** if all subsets of the set of events containg $j$ elements (for $2 \leq j \leq n-1$) are also mutually independent.

**Independence in Random Variables:** Let $X, Y, Z$ be random variables. Then $X, Y, Z$ are said to be independent if and only if:

$$P(X \in S_1 \text{ and } Y \in S_2 \text{ and } Z \in S_3) = P(X \in S_1)P(Y \in S_2)P(Z \in S_3)$$

for all possible measurable subsets $(S_1, S_2, S_3)$ of $\mathbb{R}$.

### Example

- If you roll a die 8 times, and the first 7 rolls land on a value of 1, the 8th roll is no more or less likely to roll a 5 than the first roll.
- This is true even if you roll 8 different dice which are all weighted differently provided you know nothing else about the dice (they would not be independent if you knew 7 dice favored 1 and one die favored 5, because you could use Bayes' theorem to figure it out).

Independent and identically distributed random variables are random variables which are mutually independent and have the same probability distribution (E.g. many rolls from equally weighted dice).

### Hold on a Moment!

In mathematics, a moment measures qualities of the shape of a function's graph (the set of all ordered pairs where that function is true).

The expectation of a quantity $g(X)$ for a given probability distribution is:

$$E(g(X)) \equiv \begin{cases} \sum_i g(x_i)P(X = x_i), & \text{discrete distribution} \\ \int g(x)f(x)\,dx, & \text{continuous distribution} \end{cases}$$

For a given probability distribution, the n'th moment (or the expected value of $X^n$) is $\mu'_n \equiv E(X^n)$.

We have names for some of these qualities because they are useful and interesting and useful:

- The mean ($\mu = \mu_1$) of a probability distribution is given by $\mu = E(X)$.

- The variance ($\Sigma$) of a probability distribution is given by: $\Sigma = E((X-\mu)^2)$, where the standard deviation is given by $\sigma = \sqrt{\Sigma}$.

It's important to note that not all random variables have defined analytic moments.

**The Binomial Coefficient (n choose k)**

The number of possible choices (disregarding order) of $k$ different elements from a collection of $n$ objects is a useful quantity in describing discrete probability distributions (and is even more useful in statistical and mechanical physics and will come up about as often as a simple harmoic oscillator in a similar variety of places).

If I have a collection of five marbles of different colors $\{r, b, g, y, p\}$ (for red, blue, green, yellow, and purple), the combinations of three marbles I could draw are:

$$\text{rbg}, \quad \text{rby}, \quad \text{rbp}, \quad \text{rgy}, \quad \text{rgp}, \quad \text{ryp}, \quad \text{bgy}, \quad \text{bgp}, \quad \text{byp}, \quad \text{gyp}$$

Mathematically, the way to calculate the number of different combinations available when we have $n$ objects and wish to choose $k$ of them (often referenced as n choose k) is defined:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

where $!$ indicates a factorial ($n! \equiv n \times (n-1) \times \ldots \times 2 \times 1$).

# Common Discrete Random Variables

**Binomial Random Variables**

Consider $n$ independent trials with some probability of success $p \in (0,1)$, where $X$ is the number of successes.

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n$$

$$X \sim B(n,p) \quad \leftarrow \text{Parameters (Binomial)}$$

$$\mu = E(X) = np$$

$$\text{Var}(Y) = E(Y^2 - \mu) = np(1-p)$$

**Poisson Random Variables**

A Poisson random variable counts the number of times an event occurs in a fixed interval (defined by a rate $\lambda > 0$). Consider a random variable $Y$ such that for $\lambda > 0$:

$$P(Y=y) = \frac{\lambda^y e^{-\lambda}}{y!}, \quad y = 0, 1, 2, \ldots$$

$$Y \sim \text{Poisson}(\lambda) \quad \leftarrow \text{Parameter}$$

$$\mu = E(Y) = \lambda$$

$$\text{Var}(Y) = E(Y^2 - \mu) = \lambda$$

**Geometric Random Variables**

Consider $n$ independent variables where $P \in (0,1)$. Let $X$ denote the number of failures before the first success.

$$P(X=x) = (1-p)^x p, \quad x = 0, 1, \ldots$$

$X$ is said to be geometric with parameter $p$.

$$q = 1-p, \quad E(X) = \frac{q}{p}, \quad \text{Var}(X) = \frac{q}{p^2}$$

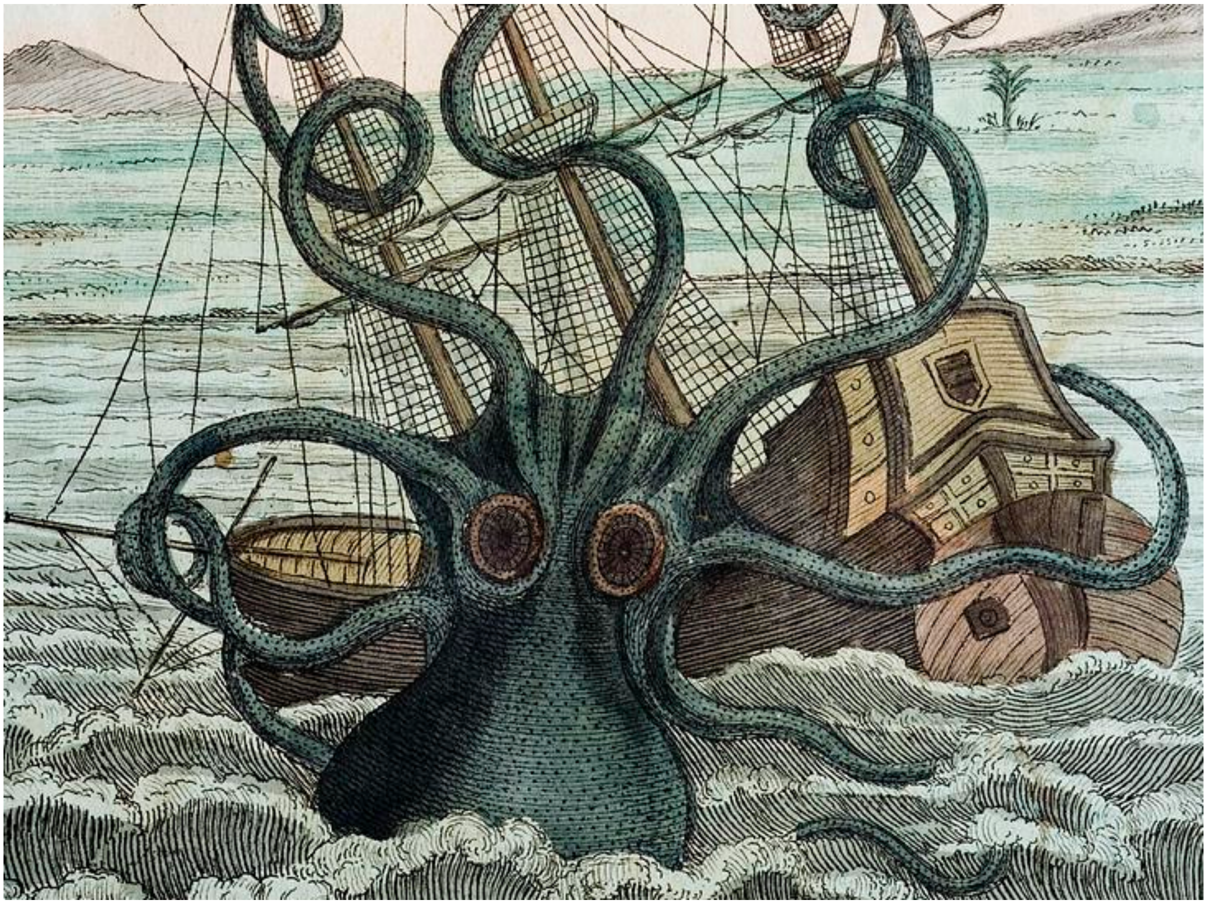
**Negative Binomial Random Variables**

Consider $n$ independent variables where $P \in (0,1)$. Let $X$ be the number of failures before $r$ successes.

$$P(X=x) = \binom{r+x-1}{x}(1-p)^x p^r, \quad x = 0, 1, \ldots$$

$$E(X) = \frac{rq}{p}, \quad \text{Var}(X) = \frac{rq}{p^2}, \quad q = 1-p$$

# Activity: Oest's Crossbow Damage



Oest was a simple grain inspector, travelling the kingdom of Minat to keep the cities free of plague. However, his ship was marooned while traveling the high seas and attacked by a kraken. With his fellow adventurers, he must defend their ship or else rest forever in Davy Jones' locker. His past experiences with vampires had left him much skill with the crossbow.

Oest's crossbow deals 3d4, rerolling ones once, and exploding. In other words, he begins by rolling three four sided dice. If he rolls a 4, he rolls an additional 4 sided die. If he rolls a 1, he rerolls that die, but if he rolls another 1, he keeps it.

The Kraken has 82 HP, and has 5 points of damage reduction (the first 5 points of damage from a given attack are ignored).

Assuming Oest does not miss any shots and scores no critical hits, What is the average number of crossbow bolts Oest must fire to slay the Kraken?

(Hint: It's impossible to solve analytically. You need to write Python code to keep track of the Kraken's HP and write a function for Oest's crossbow damage).

(Hint 2: Use recursion (see Hint 2))

# Resources

- PennState Center for Astrostatistics and Astroinformatics (online resources no longer available inspired this lecture, but they have other online resources)
- Bayesian Methods for Hackers: The book I would teach a full course out of if given the opportunity. Has Jupyter notebooks and resources online that accompany the textbook.
- Barford: Experimental Measurements: Precision, Error, and Truth is all the frequentist statistics you will need to teach experiments in modern physics.
- A good general machine learning textbook for statistical and computational methods: Hastie, Tibshirani, and Friedman: The Elements of Statistical Learning: Data Mining, Inference and Prediction
- Cornell Page on the Monte Hall Problem
- 3Blue1Brown is an excellent YouTube channel to learn about statistics and machine learning. I highly reccomend his videos on The Central Limit Theorem, and Solving Wordle using information theory.
- Maya's Notebook on Data Analysis
- Maya's Notebook on Metropolis-Hastings